

# One Learned Hamiltonian, Many Dynamics: Machine-Learning-Enabled Dissection of Which Nonadiabatic Methods Preserve Vibronic Coherence

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## Abstract

The ensemble  $S_1/S_2$  vibronic beating of a 122-atom halofluorescein heterodimer is a stringent test of nonadiabatic molecular dynamics: it is an ensemble coherence observable that exists only if the electronic coupling and its geometry dependence are captured faithfully. We train a neural quasi-diabatic Hamiltonian (NQDH) on AM1/CIS reference data and use it as a common electronic-structure surrogate for five nonadiabatic dynamics methods — mean-field Ehrenfest, fewest-switches surface hopping (FSSH), surface hopping with exact-factorization decoherence (SHXF), quantum-trajectory surface hopping (QTSH), and full multiple spawning (FMS) — at ensemble scale (300 initial conditions per method), a comparison inaccessible to direct *ab initio* dynamics. The comparison isolates a single mechanistic variable: the per-trajectory energy gate applied at transfer events. Methods that enforce it (FSSH, SHXF, FMS in both momentum-rescaling conventions) lose the beat; removing it (QTSH) restores the ensemble population balance exactly, while residual stochastic hopping dephases the coherent oscillation that only the mean-field family preserves in full.

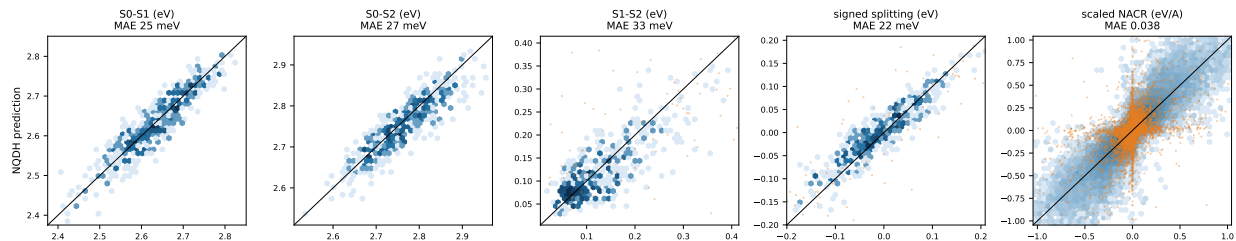


Figure 1: Model card: held-out parity of the frozen NQDH (v12a) against NEXMD references. Left to right: the two absorption gaps, the dynamics gap, the signed diabatic splitting, and the scaled-NACR components. Blue density: thermal test set; orange: displaced (post-excitation) geometries. *TODO: final stats in caption.*

## 1 Introduction

## 2 Methods

### 2.1 The neural quasi-diabatic Hamiltonian

### 2.2 Dynamics

## 3 Results

### 3.1 The learned Hamiltonian reproduces the vibronic beat

### 3.2 The energy gate decides survival of the beat

### 3.3 When the gate closes: timing of frustrated events

## 4 Analysis

## 5 Discussion

## 6 Proposed directions

## 7 Conclusions

## Code and data availability

All code, the frozen model, initial conditions, and per-figure data/scripts are available in this repository (<https://github.com/vmfreixas/nqdh-beating-demo>).

## Acknowledgments

## References

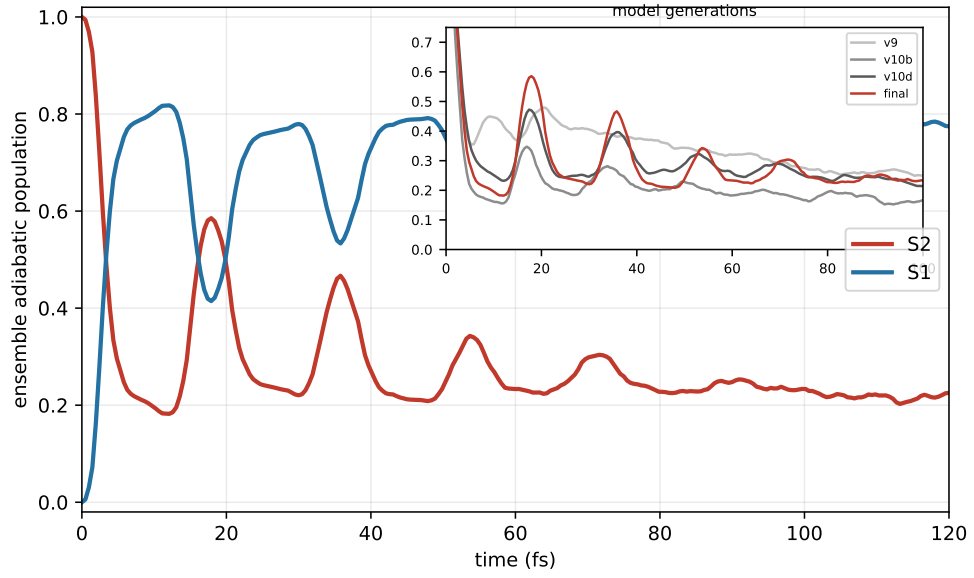


Figure 2: The  $S_1/S_2$  vibronic beat from 300-trajectory diabatic Ehrenfest ensembles on the frozen NQDH: recurrences every  $\sim 18$  fs decohering by  $\sim 90$  fs, in agreement with the reference NEXMD phenomenology. Inset: progression across model generations. *TODO: annotate recurrence times.*